



Cloud & Edge Computing

Workshop AFC 2026

Embrace the **future of IT** today.

Tom Claes - CSA Azure Infra at Microsoft

Laurens Giesen – Student Information Management / Intern Engineering Studio at Cegeka



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3   "age": 27,
4   "current_role": {
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Who are you?

MS/BA of AI?

Master of
Information
Management?

MS/BA of Business
Economics?

Master of Applied
Informatics?

MS/BA of Industrial
Engineering?

MS/BA of Computer
Science?

MS/BA of Business
Engineering?

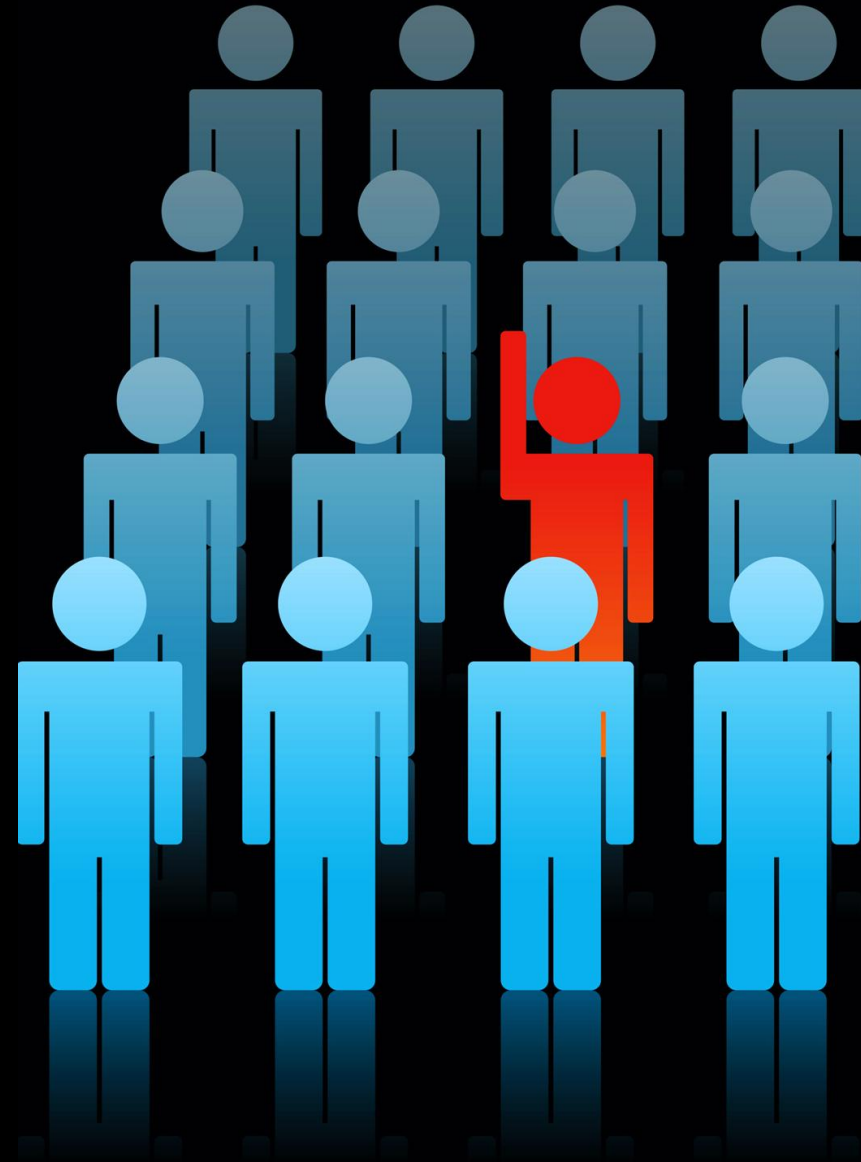
Ground rules

Let's keep it as **interactive** as possible

Don't hesitate to interrupt me to ask **questions** during the presentation

If not feeling comfortable, please ask me your questions during the break / after the event

We will have **breaks** during the event to go for drinks / to go to the toilet 😊



Introduction to Cloud Computing

Introduction to Azure Cloud Platform

Computing capabilities

Product portfolio

Fundamentals

Introduction to Azure OpenAI

Break 10'

Edge computing

Introduction to Containerization & k8s

Use case Laurens at Cegeka

Break 10'

Demo: **Microsoft Docs AI assistant with Azure Container Apps and Azure OpenAI**

- **Local dev + creation of Dockerfile and Docker image**
- **Setup Azure environment using Azure CLI**
- **Push to Azure Container Registry**
- **Deploy application to Azure Container Apps**

Agenda

Objectives

Why are you here?

Familiarity with important concepts

Industry relevance

Hands-on project using Azure & containers

Out of scope

Become an expert in all the topics

Question

Familiarity with the Cloud?

What is the preferred Cloud Provider?

Explain the cloud in one sentence



What is the cloud?

An introduction to cloud computing
with Microsoft Azure

Presenter: **Clay Baker**

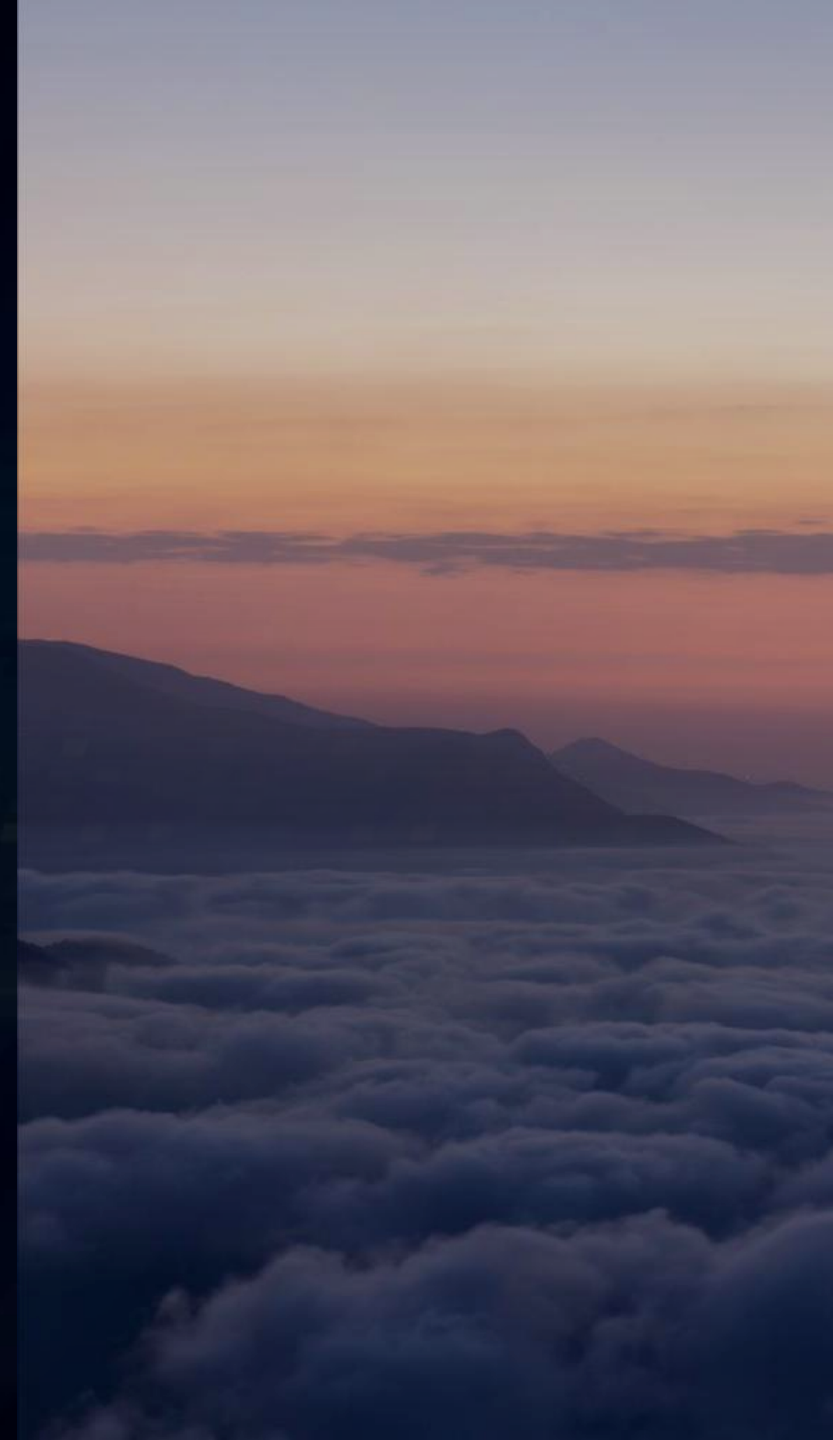
[@claybaker](#)



Introduction to Cloud Computing

“Simply put, cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet to offer faster innovation, flexible resources, and economies of scale.”

- Capabilities are hosted at a remote data center managed by a CSP
 - Compute
 - Storage
 - Networking
 - Development tools
- Well known CSPs (Cloud Solution Providers)
 - Azure
 - GCP
 - AWS



Advantages of the Cloud

"Cloud computing is a big shift from the traditional way businesses think about IT resources."

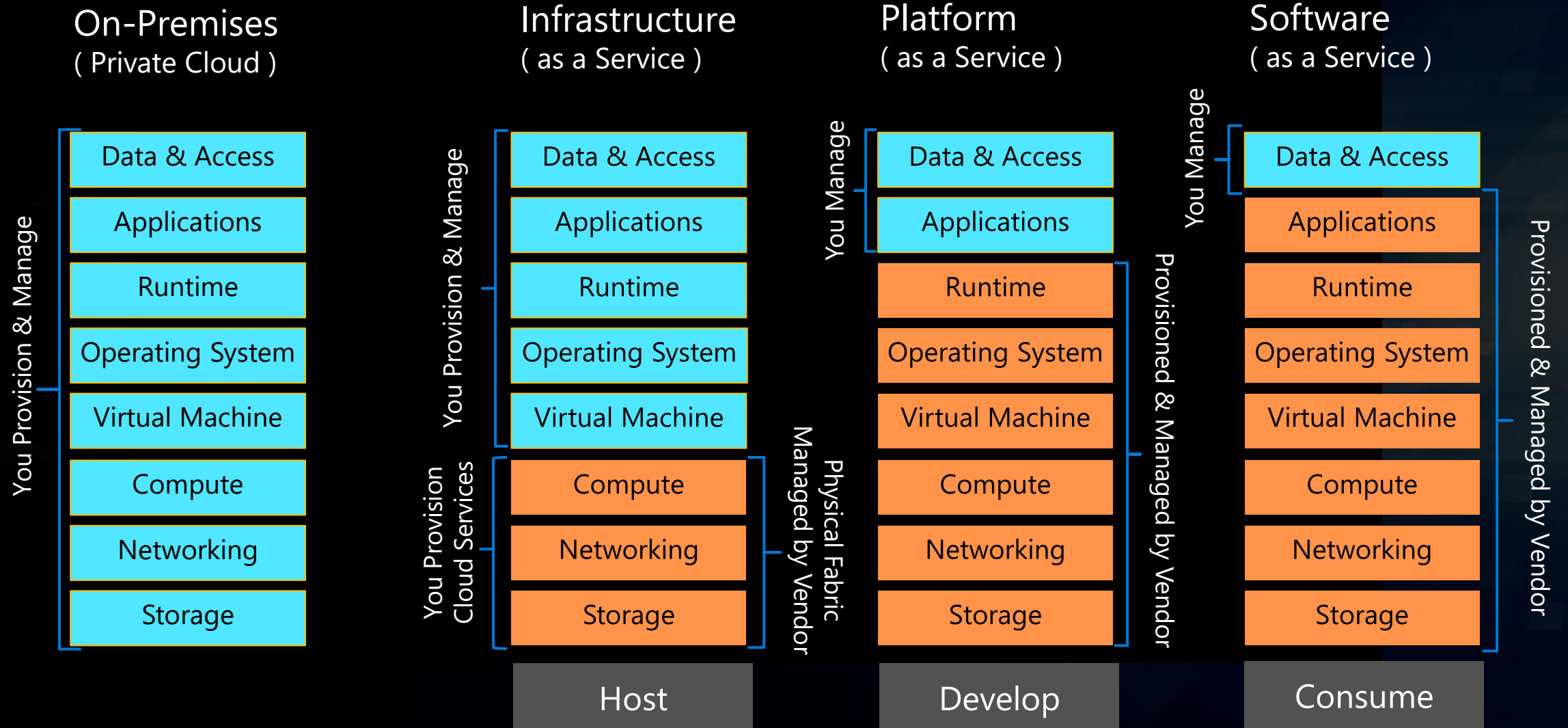
- Optimization of costs
 - CapEx to OpEx
- Speed of services
 - Provisioning of compute in minutes
- Ability to scale elastically
- Focus on core business, less on time-consuming IT management
- Performance of services
- Reliability
- Security



Questions?



Azure Cloud Computing Capabilities



Platform Services

Security & Management

- Portal
- Azure Active Directory
- Azure AD B2C
- Multi-Factor Authentication
- Automation
- Scheduler
- Key Vault
- Store/ Marketplace
- VM Image Gallery & VM Depot

Services Compute

- Cloud Services
- Service Fabric
- Batch
- RemoteApp

Integration

- Storage Queues
- BizTalk Services
- Hybrid Connections
- Service Bus

Media & CDN

- Media Services
- Content Delivery Network (CDN)

Web and Mobile

- Web Apps
- API Apps
- Mobile Apps
- Logic Apps
- API Management
- Notification Hubs

Developer Services

- Visual Studio
- Azure SDK
- VS Online
- App Insights

Data

- SQL Database
- Data Warehouse
- DocumentDB
- Redis Cache
- Azure Search
- Storage Tables

Analytics & IoT

- HDInsight
- Machine Learning
- Stream Analytics
- Data Lake
- Data Factory
- Event Hubs
- Data Catalog
- IoT Hub
- Mobile Engagement

Hybrid Operations

- Azure AD Health Monitoring
- AD Privileged Identity Management
- Domain Services
- Backup
- Operational Analytics
- Import/Export
- Azure Site Recovery
- StorSimple

Infrastructure Services

OS/Server Compute

- Virtual Machines
- Container Service

Storage

- BLOB Storage
- Azure Files
- Premium Storage

Networking

- Virtual Network
- Load Balancer
- DNS
- Express Route
- Traffic Manager
- VPN Gateway
- App Gateway

Datacenter Infrastructure (54 Regions, 100+ Datacenters)



Important concepts

✓ Automation

IaC
CI/CD
Policy/Rule enforcement

➔ Reduce manual operational tasks as much as possible

✓ Operational models

IaaS vs PaaS vs SaaS
Serverless

✓ Resource organization

Landing Zones

✓ Security

🔒 Don't trust anyone / zero-trust / expect to be compromised

✓ High Availability and Disaster Recovery

✓ Cloud Sovereignty

✓ (Private) Networking

✓ Cost awareness & FinOps

✓ Observability

✓ ...

ABOUT

The Backbone of Microsoft Cloud: Our Datacenters

 Explore the globe

Microsoft is continuing to make substantial investments in [datacenter](#) infrastructure to support the growing demand for artificial intelligence (AI) and cloud services. This year, we're investing \$80 billion toward the development of AI-enabled datacenters around the world.

70+	Azure regions	400+	Datacenters worldwide
600k+	Kilometers of network	190+	Network PoPs



Data Center video

AZURE BELGIUM CENTRAL

TRANSFORM

INNOVATION

**Microsoft opens its first
cloud region in Belgium
accelerating innovation
and economic growth**



Loutfi Belghmidi
PR & Comms Lead Microsoft Belux

[Microsoft opens its first cloud region in Belgium accelerating innovation and economic growth – Microsoft Pulse](#)

Fun facts

Fiber

HQ

Size for reference

Fun locations

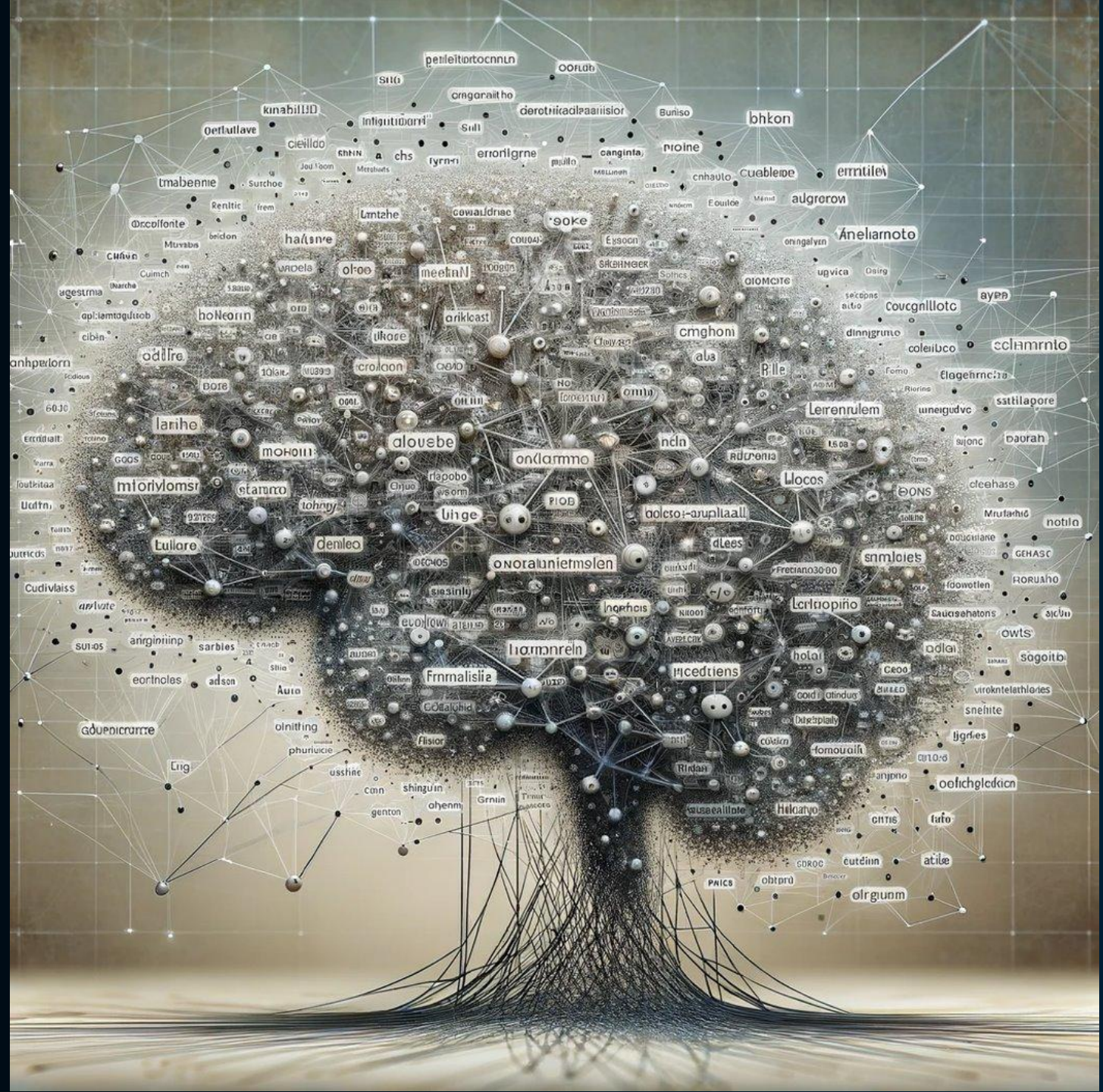


Questions?





AI / LLM



Artificial Intelligence

Machine Learning



Generative AI



Artificial Intelligence

The field of computer science that seeks to create intelligent machines that can replicate or exceed human intelligence



Machine Learning

Subset of AI that enables machines to learn from existing data and improve upon that data to make decisions or predictions

IBMs supercomputer Deep Blue vs Garry Kasparov



Deep Learning

A machine learning technique in which layers of neural networks are used to process data and make decisions

GPU became available: One example is AlexNet, a CNN



Generative AI

Create new written, visual, and auditory content given prompts or existing data

- 2014 GAN: Generative Adversarial Network, imitating pictures
- 2022: ChatGPT, combination Generative AI & NLP

What does ChatGPT stand for?

Chat Generative Pre-Trained Transformer

Attention is all you need.

Large Language Foundation models

Microsoft AI at Scale

Project Turing

17B
Turing-NLG | 2020

340M
BERT-Large | 2018



Our mission is to expand the boundaries of natural language understanding.

machine reading comprehension, question answering, transfer learning, reinforcement learning, computer vision, and even building interpretable models

Turing-NLG

Megatron NLG Turing

T-ULRv5

DeepSpeed

ZeRO

175B
GPT-3 | 2020

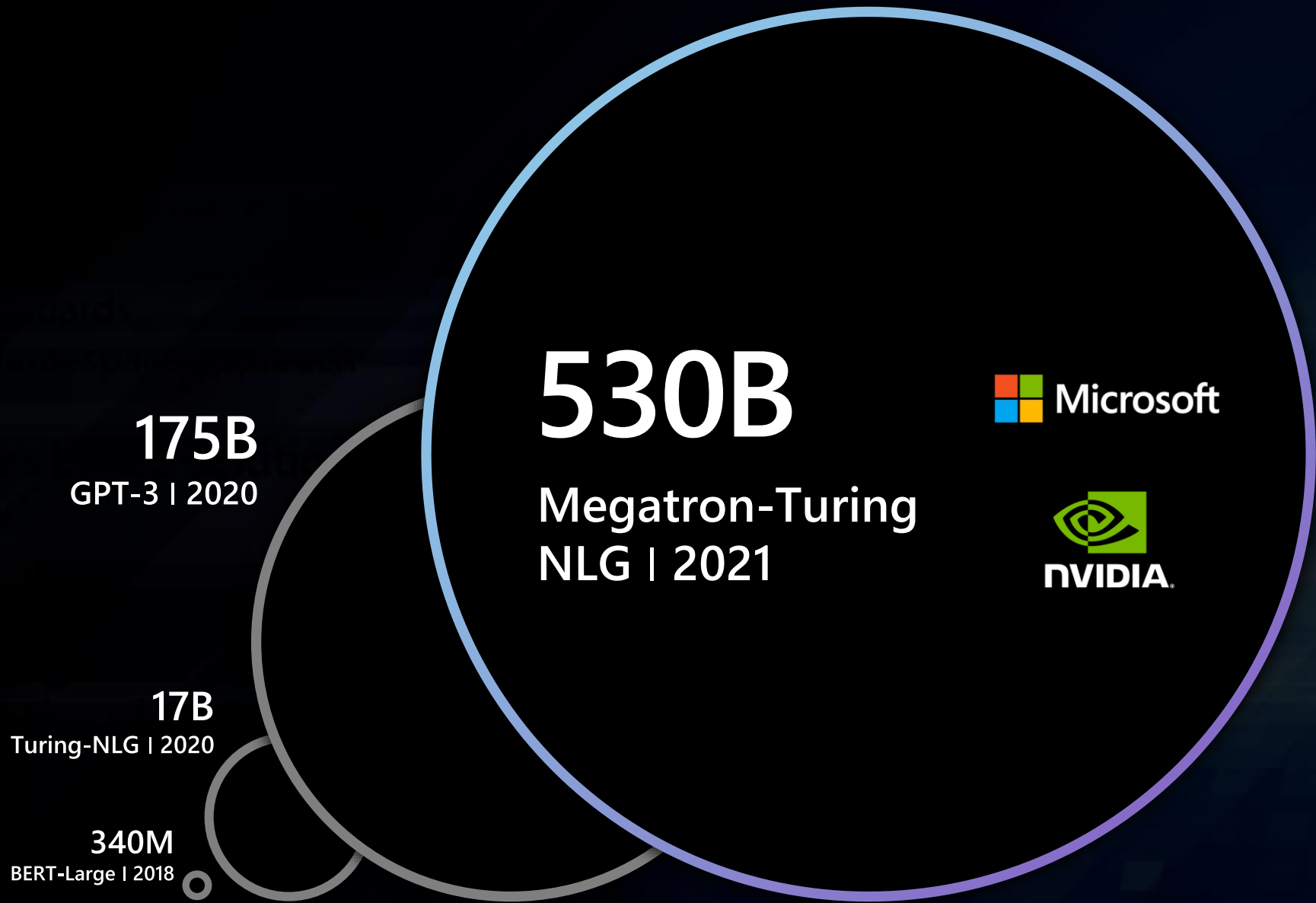
17B
Turing-NLG | 2020

340M
BERT-Large | 2018



OpenAI





175B
GPT-3 | 2020

17B
Turing-NLG | 2020

340M
BERT-Large | 2018

530B

Megatron-Turing
NLG | 2021



+ -1.7-1.8 TRILLION
GPT 5

175B
GPT-3 |
2020

17B
Turing-
NLG | 2020

340M
BERT-Large | 2018

530B
Megatron-Turing
NLG | 2021

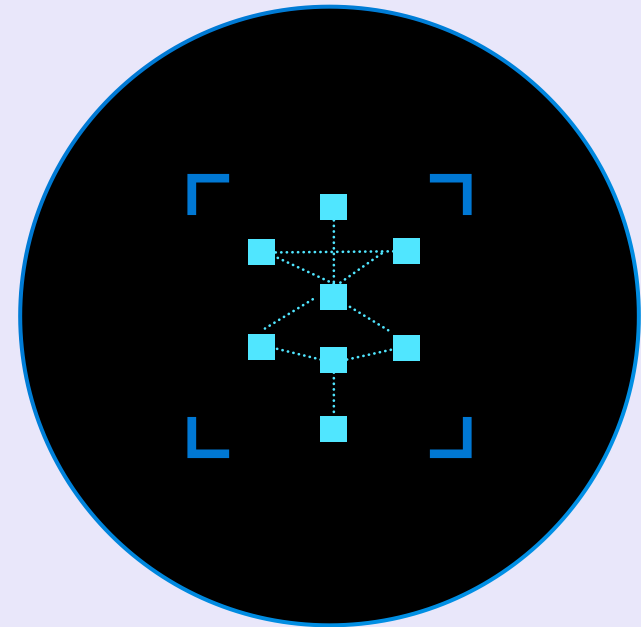
Embeddings

An embedding is a special format of data representation that can be easily utilized by machine learning models and algorithms.

The embedding is an information dense representation of the semantic meaning of a piece of text.

Each embedding is a vector of floating-point numbers, such that the distance between two embeddings in the vector space is correlated with semantic similarity between two inputs in the original format.

For example, if two texts are similar, then their vector representations should also be similar.



Embeddings make it possible to map content to a “semantic space”

A neutron star is the collapsed core of a massive supergiant star



[15 34 24 13 ...]

A star shines for most of its active life due to thermonuclear fusion

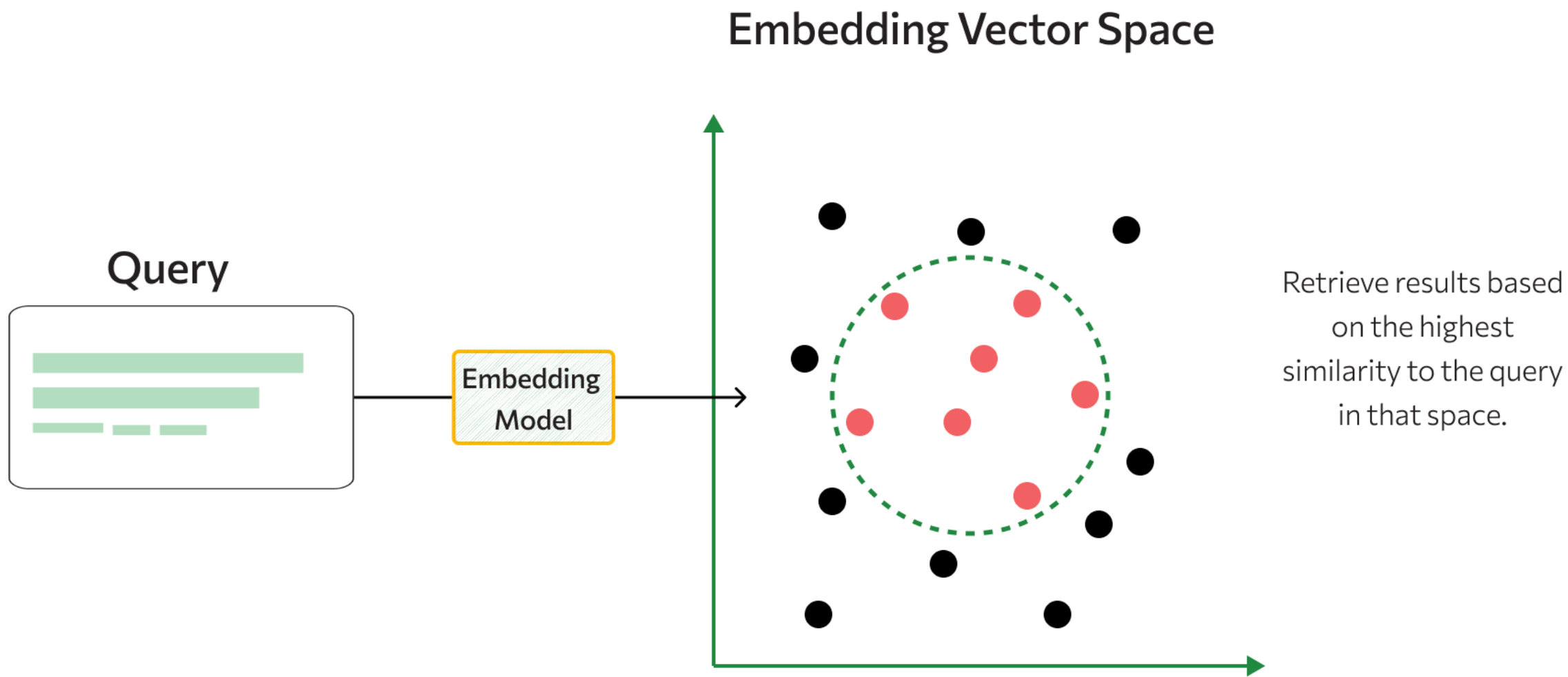


[16 22 89 26 ...]

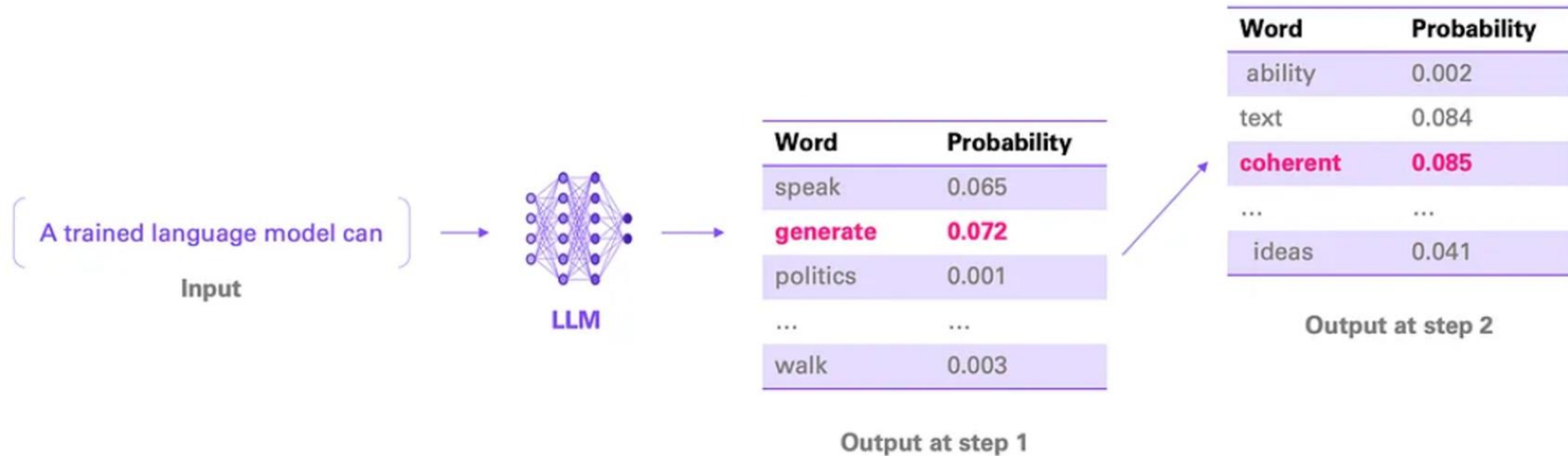
The presence of a black hole can be inferred through its interaction with other matter



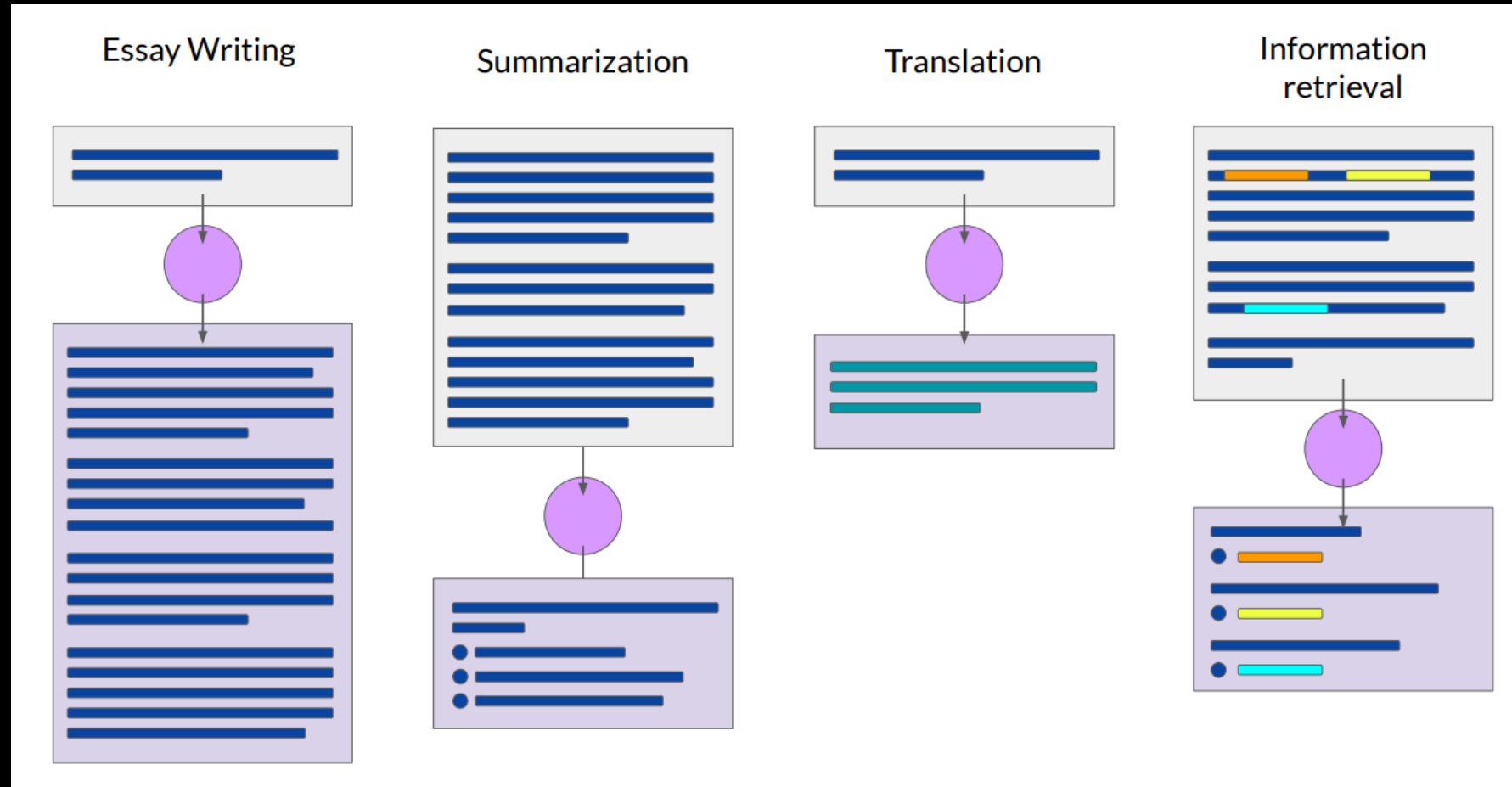
[20 13 31 89 ...]



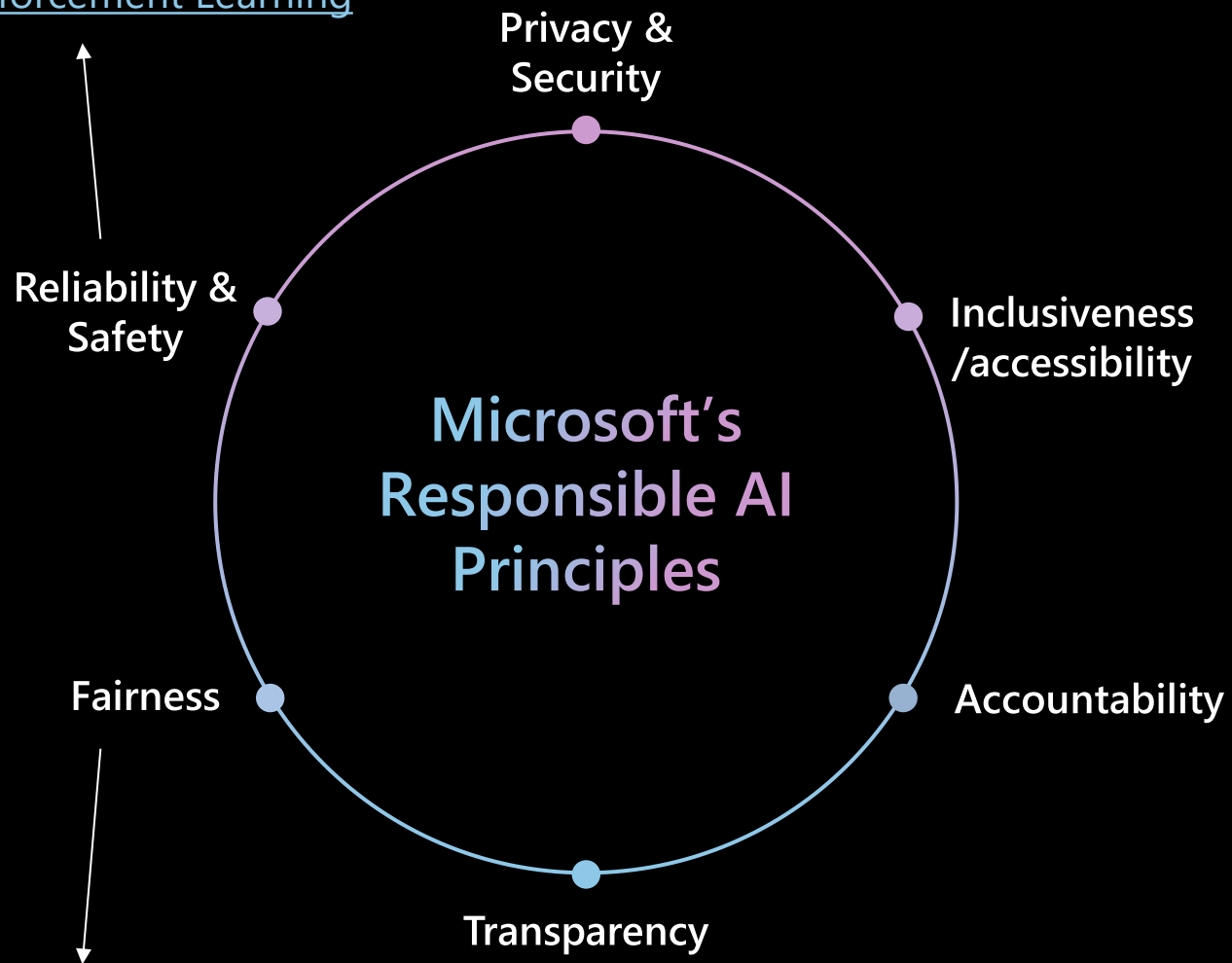
Generate text by predicting one word at the time



Tasks & Use cases



Tackling Snow-Induced
Challenges: Safe Autonomous
Lane-Keeping with Robust
Reinforcement Learning



Building blocks to enact principles



Tools and processes



Training and practices



Rules



Governance

Google Translate is biased, and
here's how we can fix it!

Azure AI

Applications



Partner Solutions

Application Platform

AI Builder



Power BI



Power Apps



Power Automate



Power Virtual Agents

Scenario-Based Services

Applied AI Services



Bot Service



Cognitive Search



Form Recognizer



Video Indexer



Metrics Advisor



Immersive Reader

Customizable AI Models

Cognitive Services



Vision



Speech



Language



Decision

Azure OpenAI Service

ML Platform



Azure Machine Learning

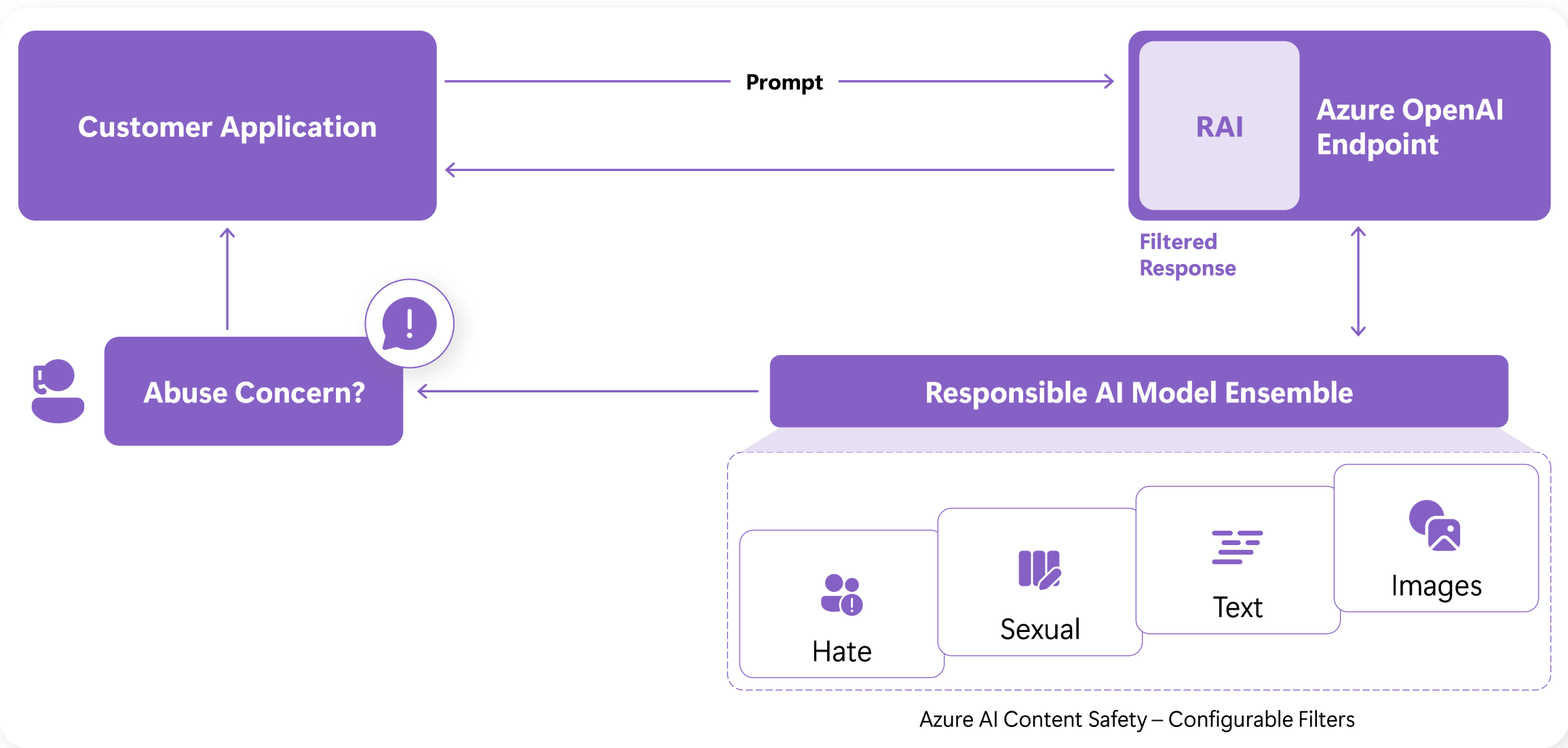


Business Users



Developers & Data Scientists

Responsible AI in Azure OpenAI Service



MCP – what's that?

- Open protocol that lets an AI app connect to external tools and data (APIs, databases, files, services)
- MCP uses a client–server model: an AI “host” connects to one or more MCP servers that expose capabilities (“tools”)
- Why it matters for LLMs: LLMs don't automatically know **your live data or your systems** - MCP helps them retrieve fresh info and take actions using approved tools.
- Without MCP, every AI tool needs custom integrations for every system; MCP makes it more plug-and-play across different clients and servers
- Security & control: access can be governed via authentication/authorization and controlled exposure of tools

Examples:

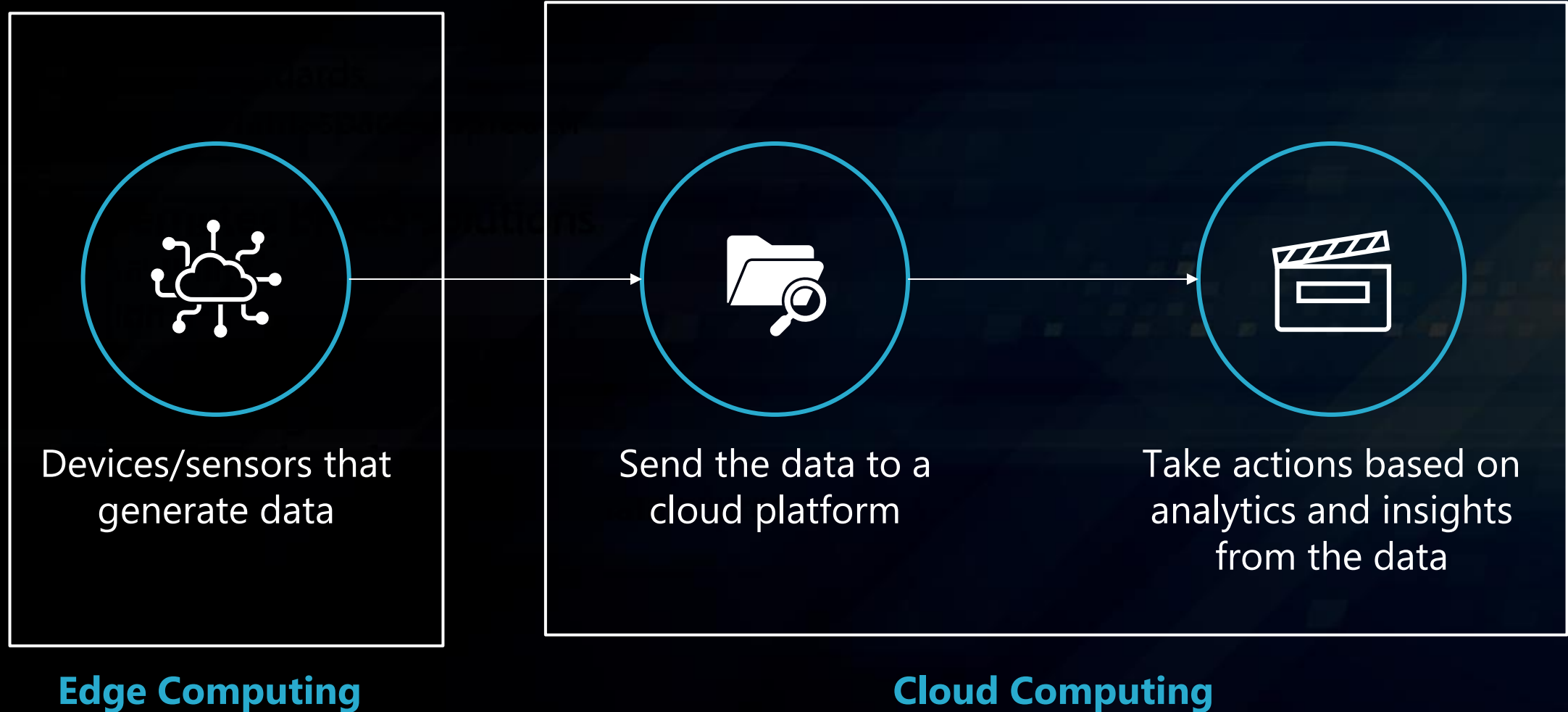
- Microsoft Learn MCP Server
- Azure MCP Server
- KU Leuven Toledo MCP Server (?)

10 minutes break

Edge Computing



Internet of Things - overview





Introduction to Edge Computing



Introduction to Edge Computing

"Edge computing is a distributed computing framework that allows IoT devices to quickly process and act on data at the edge of the network, in a decentralized manner."

- Solves three interrelated challenges:
 - Connecting a device to a network from a remote location
 - Desert, ocean...
 - Slow data processing due to network or computing limitations
 - Edge devices causing network bandwidth issue
 - E.g., security cameras only sending relevant data to the cloud
- Use cases:
 - Autonomous driving
 - Vessels (remote areas)
 - Security cameras
 - Agriculture



Benefits of Edge Computing

"From workplace safety to security and productivity, the benefits of edge computing are vast."

- More efficient operations
 - Day-to-day operations
- Faster response times, reduced latency
- Improved workplace safety
- Reduced IT costs
 - Less data sent to the cloud
 - Reduced bandwidth
- Increased reliability
 - Reducing reliance on a central cloud infrastructure
- Ability to operate in remote areas
 - Deserts
 - Oceans
 - Farming in rural areas



Future of IoT (at Microsoft)

- Focus on Open Source & Industry standards
 - MQTT protocol
 - ISA 95 standards
 - Unified Namespace approach
- Kubernetes based solutions
 - Scalability
 - High availability
- AI at the Edge
 - Cloud: training of models
 - Edge: inferencing close to the data source

Edge AI

Computer vision at the Edge

- Safety/security cameras
- Autonomous driving

Agriculture: anomaly detection and monitor livestock in rural areas

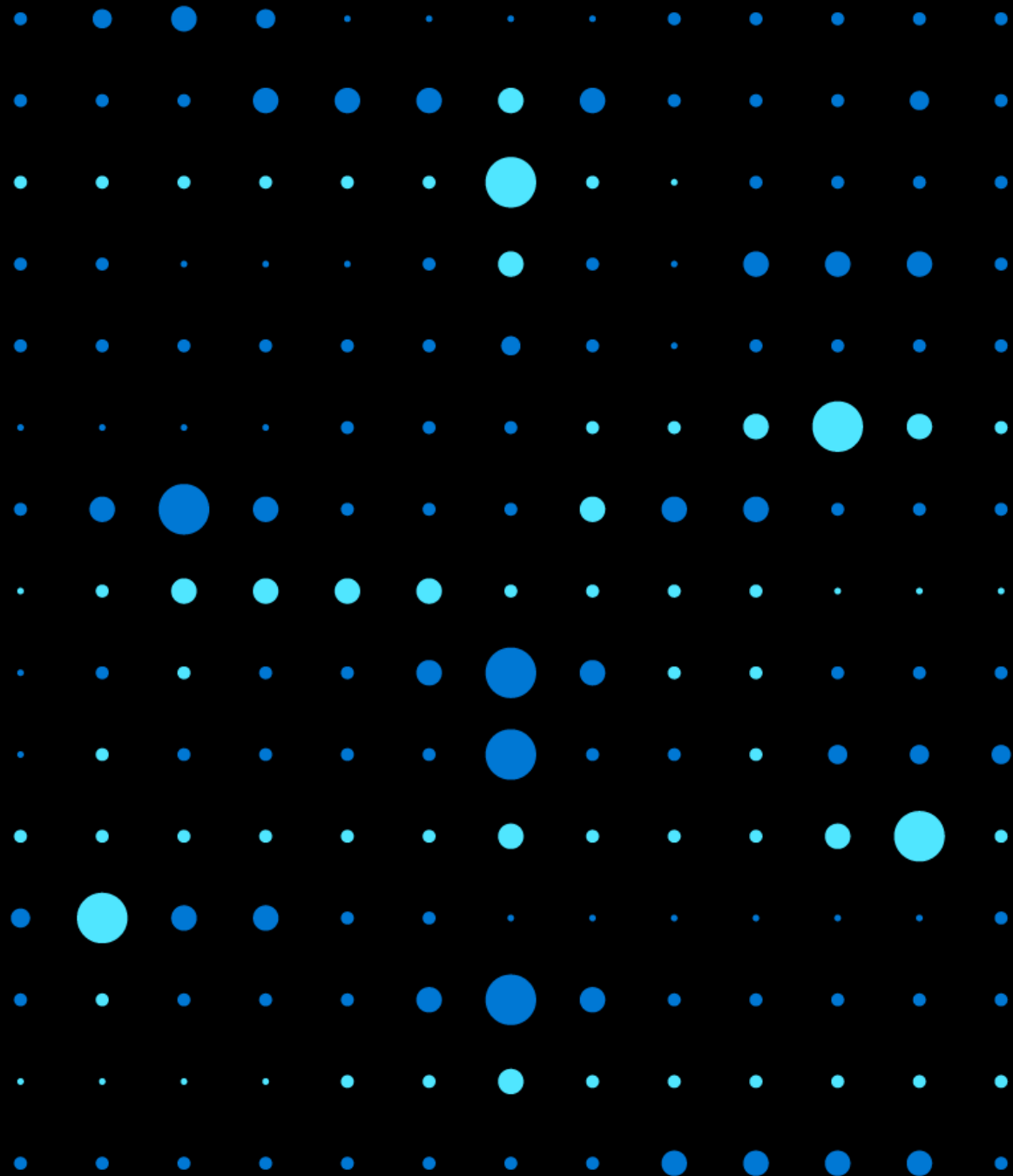
Smart apps:

- Chatbot at the edge with out internet connection
- Predictive maintenance in the factories

Questions?



Containerization and Kubernetes



How to host application on the cloud



Control

Productivity



VMs

Azure Virtual Machines

Containers

Azure Kubernetes Service | Azure
Red Hat OpenShift

PaaS/Serverless

App Service | Functions |
Spring Apps | Container Apps

Low code

Power Apps | AI Builder
Azure Services

Managed databases

Azure SQL Database | PostgreSQL | MySQL | Cosmos DB

DevOps

GitHub | Azure DevOps | VS Code

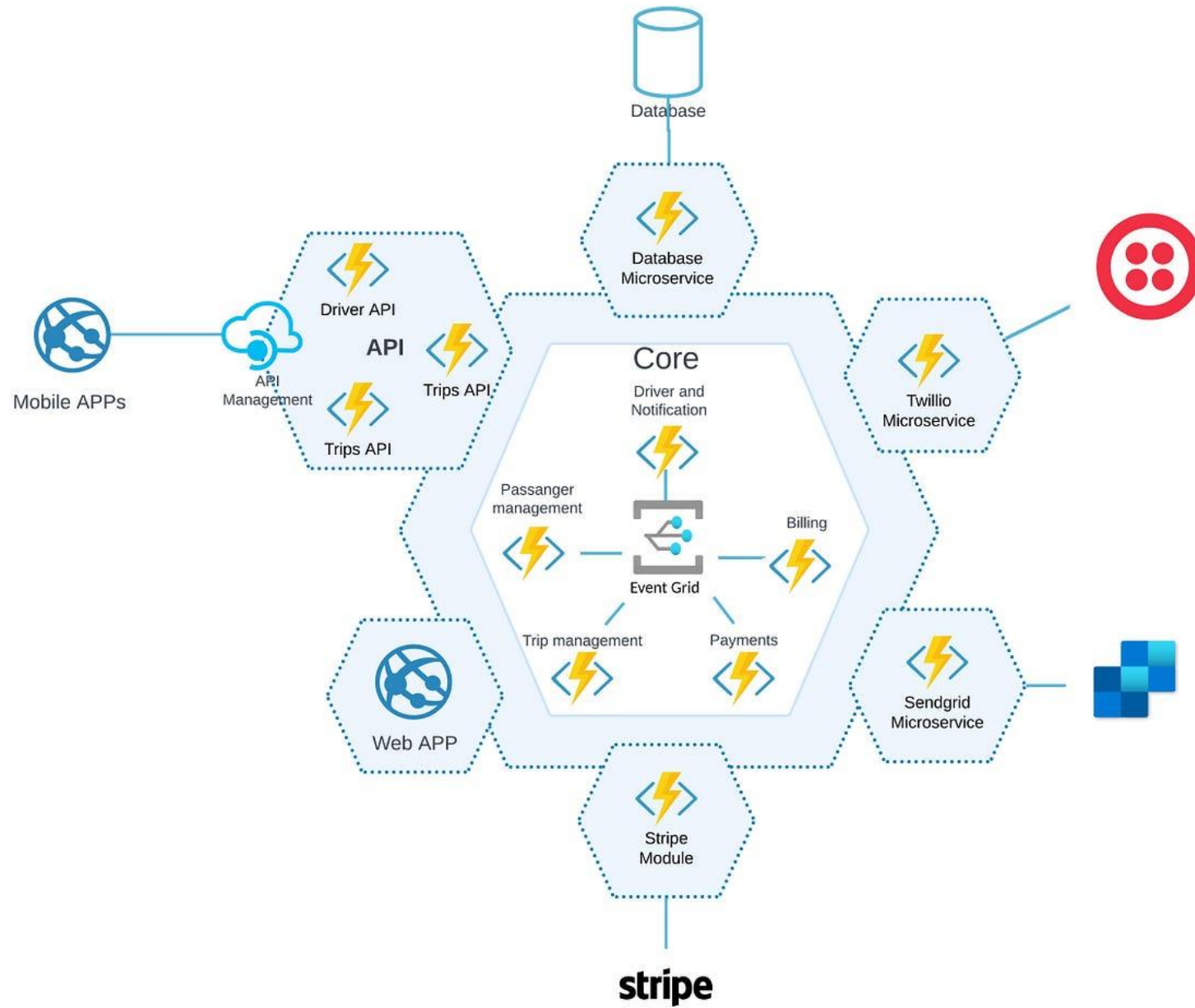


Azure | Azure Arc | Azure Stack

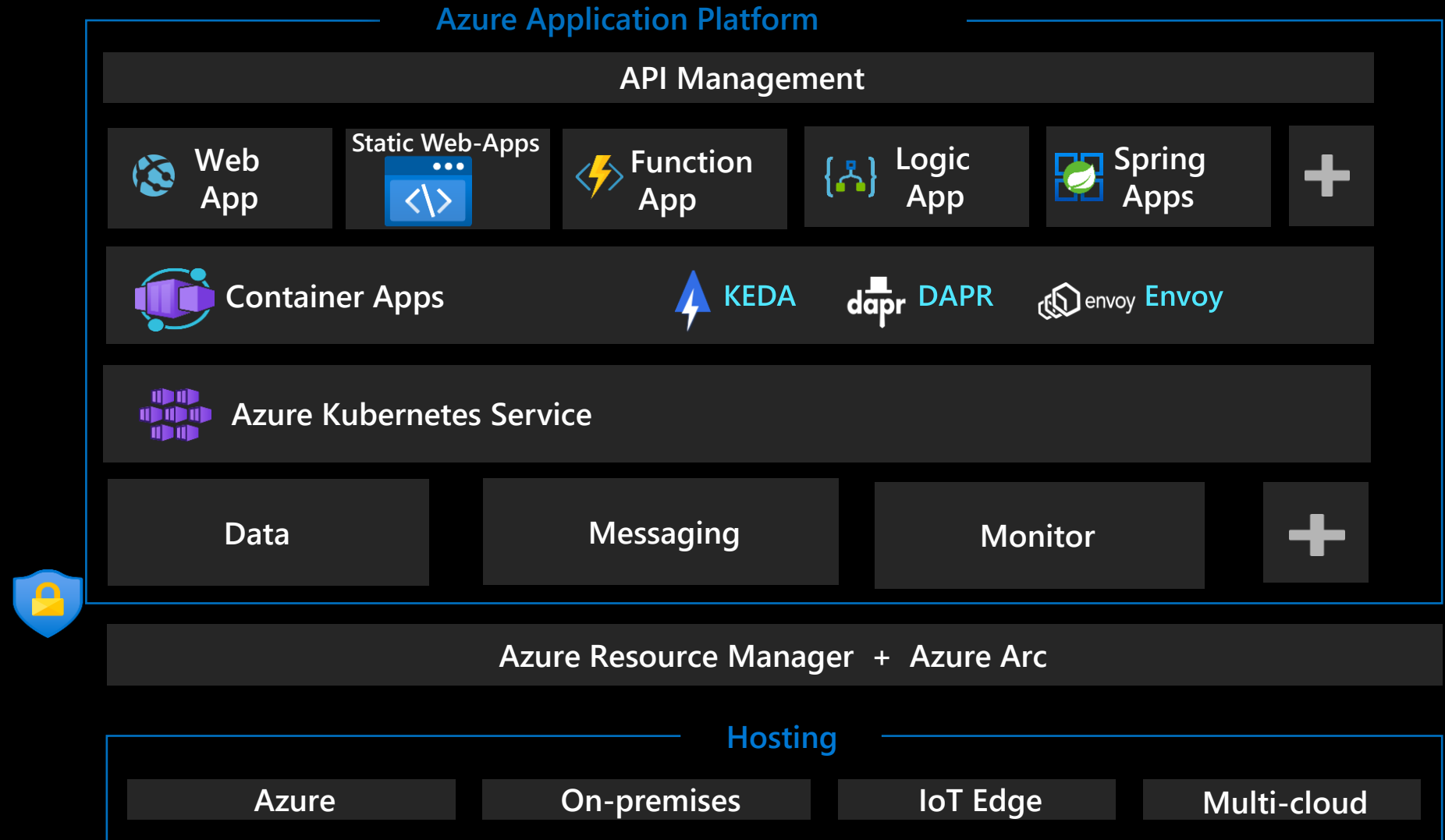


Microservices

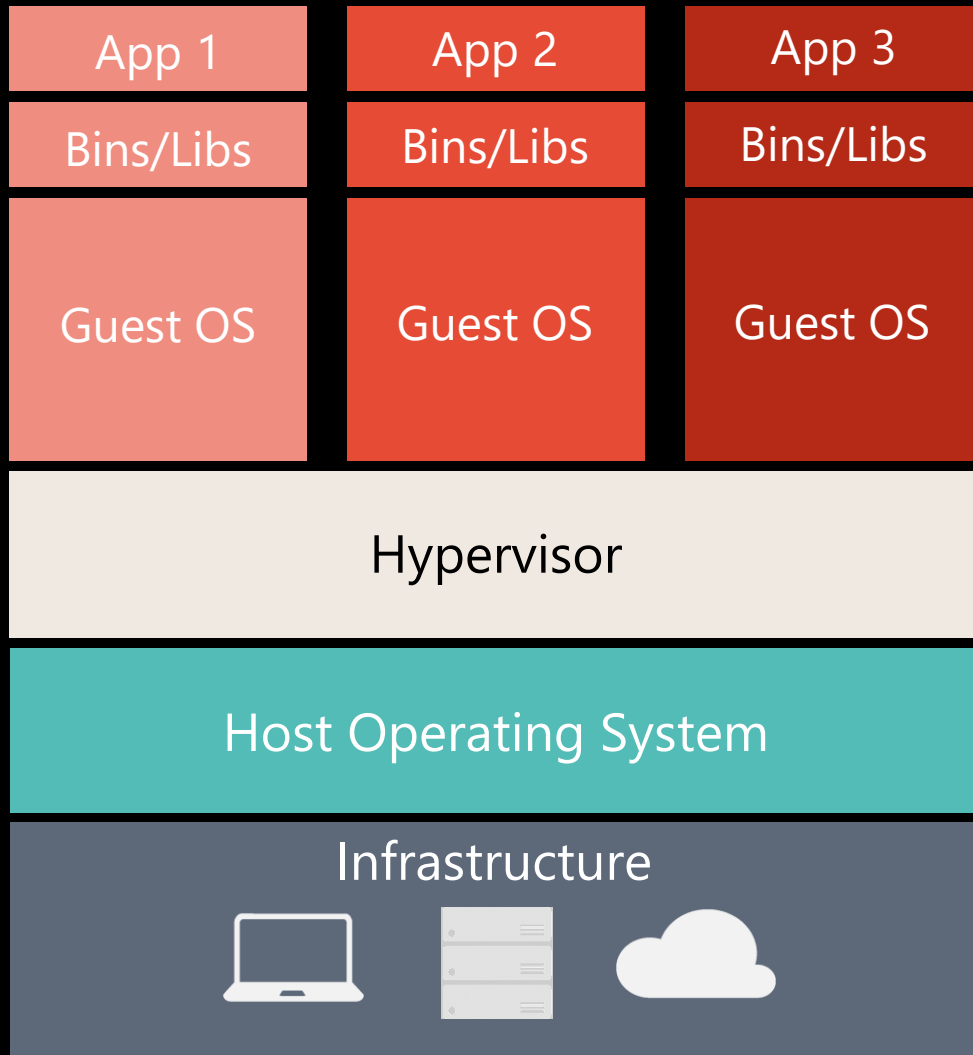
- Architectural framework
- Decentralized management and development
- Technology & language agnostic
- Scalability & elasticity



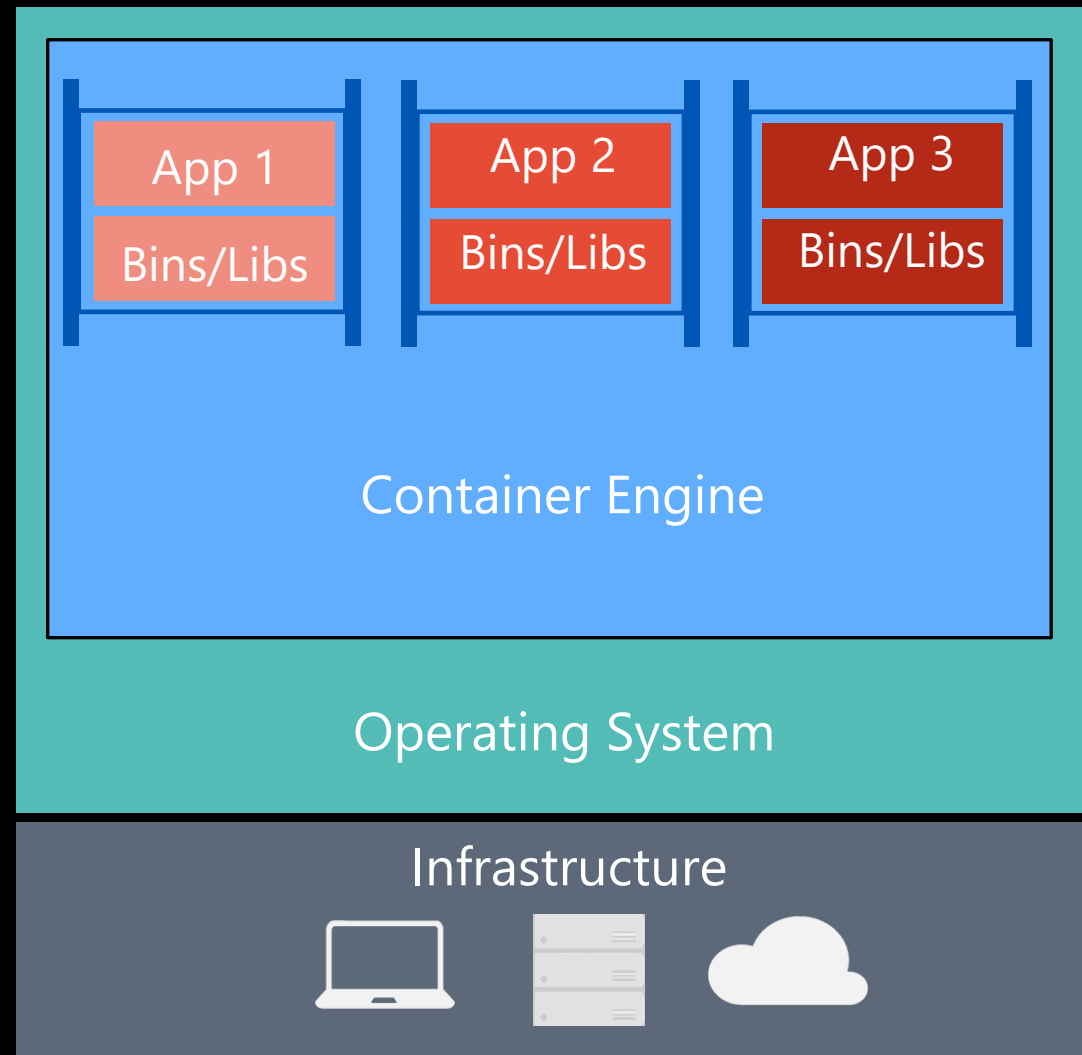
Azure's Application Platform



Virtual Machines



Containers



Why containers?

- Resource Efficiency
- Portability
- Faster Startup Times
- Simplified Management
- Immutability and Version Control
- Agile development of applications
- Microservices architecture

Kubernetes: the de facto container orchestrator

Microsoft investing in Kubernetes

88%

of enterprises using or adopting Kubernetes

- 74% of enterprises that use Kubernetes are running workloads in production
- 74% of survey respondents have a DevSecOps initiative underway
- At least 20% of enterprises that use Kubernetes leverage **six different open-source security tools**



Portable

Public, private, hybrid, multi-cloud

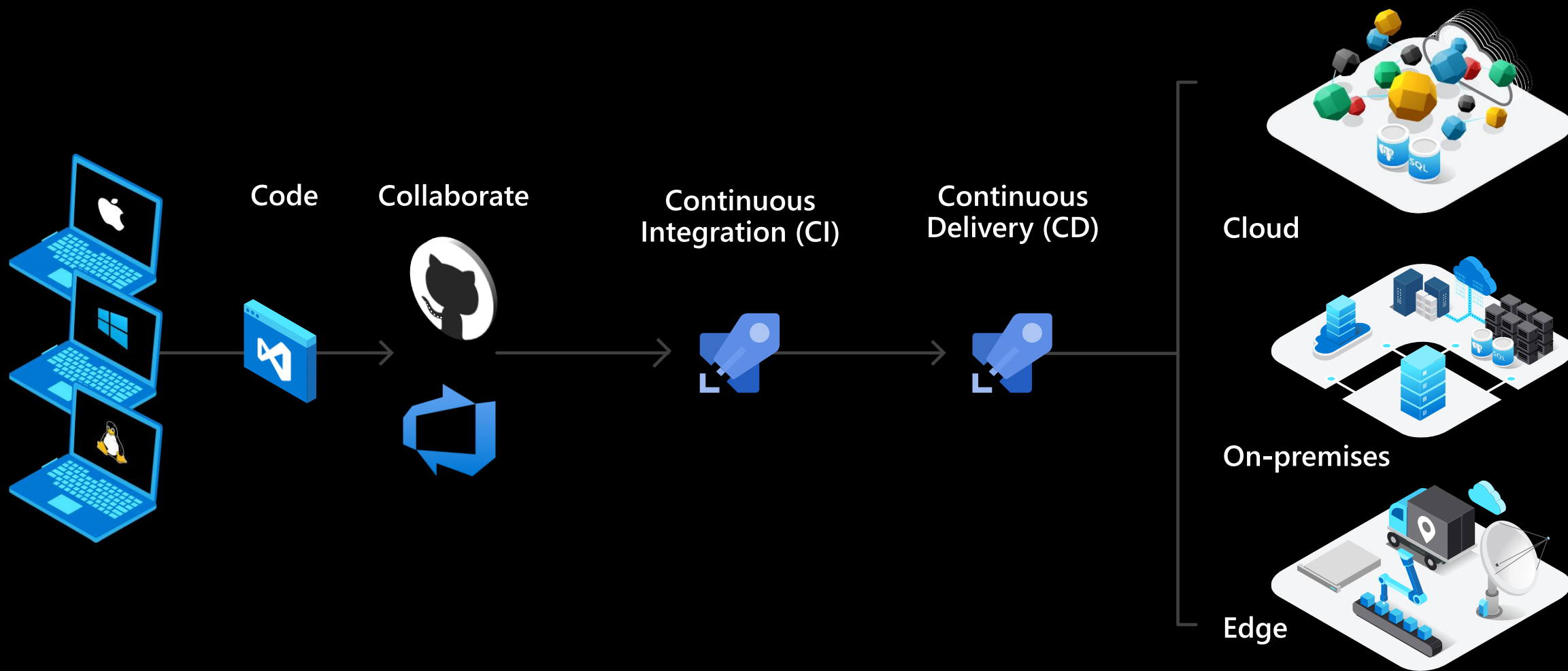
Extensible

Modular, pluggable, hookable, composable

Self-healing

Auto-placement, auto-restart, auto-replication, auto-scaling

Code to Cloud

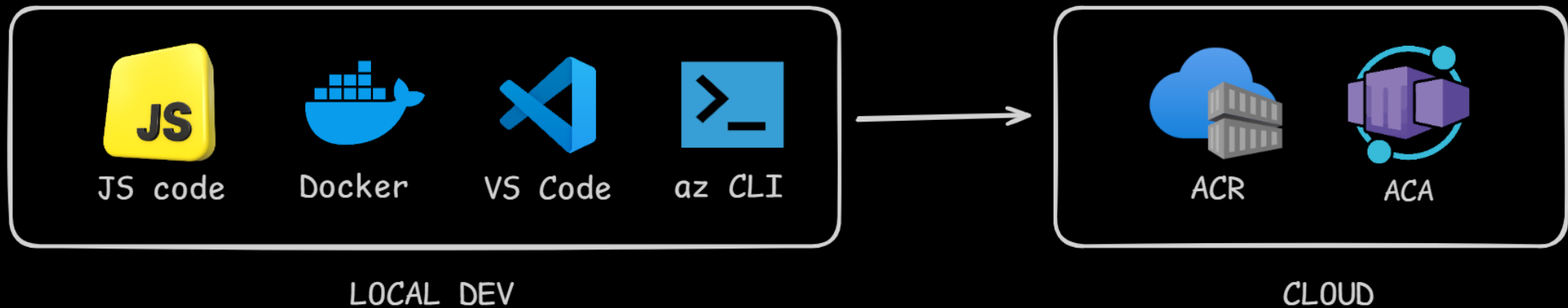


Use case Laurens

Questions?



Demo



Want to build it yourself?

→ [Build your own Microsoft Docs AI assistant with Azure Container Apps and Azure OpenAI | All things Azure](#)

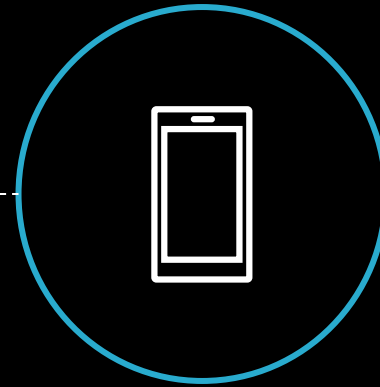
Recap of the evening



Cloud Computing



Azure (OpenAI)

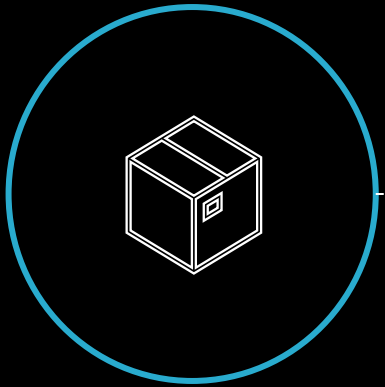


Edge Computing



Azure (IoT) Platform

Recap of the evening



Containerization &
AKS



Use case at
Cegeka



Hands-on demo



Next steps

Feel free to connect over LinkedIn!



Tom Claes



Laurens Giesen

Recommended learnings

Azure Fundamentals certifications:

- AI: [Microsoft Certified: Azure AI Fundamentals - Certifications | Microsoft Learn](#)
- Data: [Microsoft Certified: Azure Data Fundamentals - Certifications | Microsoft Learn](#)
- Azure infra: [Microsoft Certified: Azure Fundamentals - Certifications | Microsoft Learn](#)
- Security: [Microsoft Certified: Security, Compliance, and Identity Fundamentals - Certifications | Microsoft Learn](#)

- ✓ How to get certificates at a discounted price
- ✓ How to set up your Azure environment and get free credits (100\$/year)

➔ [Student Certifications - Student Hub | Microsoft Learn](#)

Overview of benefits for students

➔ [Student Hub Overview - Microsoft Learn Student Hub | Microsoft Learn](#)